OAK INTERNATIONAL SCHOOL

MYP 5 MATHEMATICS — MID-TERM TEST 2

Bilingual Excellence in Education & Leadership

DURATION	30 minutes
TOTAL MARKS	20
CALCULATOR	Not permitted

Name: _____ **Class:** _____ **Date:** _____

Answer ALL questions. Show all working — marks are awarded for method, for general rules, for verification of rules, and for interpretation in context. Write final answers on the dotted lines. Give non-exact decimal answers correct to 2 d.p. Amounts in francs are exact. Aim for about 90 seconds per mark.

TASK 1 — UNDERSTANDING NUMBER PATTERNS — Criterion A • 4 MARKS

For a class display, three students each recorded a number pattern:

Ali: 4, 10, 16, 22, … **Bea:** 1, 4, 9, 16, … **Camille:** 40, 36, 32, 28, …

(a) Two of the three patterns change by a constant amount. Identify these two patterns and state the constant change in each.

[2 marks]

(b) A classmate says that Bea's pattern is linear *"because it always increases."* Explain the classmate's error, using calculations.

[2 marks]

TASK 2 — INVESTIGATION: STONE-EDGED GARDEN BEDS — Criterion B • 6 MARKS

For the school's greening project, the mathematics club builds triangular garden beds edged with equal stones. Beds 1, 2 and 3 are shown, in order.



(a) Determine the number of stones needed for Bed 4 and Bed 5.

[2 marks]

(b) The club describes the pattern with a general rule. Write a formula for the number of stones, S, in Bed n, where n is the bed number.

[2 marks]

(c) The gardener has ordered 30 stones — according to the club, exactly enough for Bed 10. Use your rule to verify the order, and explain why your rule must work for *every* bed, referring to the structure of the pattern rather than only to the numbers.

[2 marks]

TASK 3 — APPLICATION: AMINA'S SAVINGS CHALLENGE — Criterion D • 6 MARKS

Amina joins a one-year savings challenge in Cotonou. She deposits 10 000 francs in January (month 1), and from then on she increases each month's deposit by exactly 2 500 francs.

(a) Show that Amina deposits $(2\,500n + 7\,500)$ francs in month n.

[2 marks]

(b) How much does Amina deposit in December (month 12)?

[1 mark]

(c) Calculate the total amount Amina has saved by the end of December.

[2 marks]

(d) In January, Amina sees a bicycle costing 300 000 francs, payable at the end of December. Determine whether her savings will be enough. Justify with a calculation.

[1 mark]

TASK 4 — REASONING CHALLENGE: TWO CANDLES — Criterion A • 4 MARKS

Two candles are lit at the same moment. Candle A is 24 cm tall and burns 1.5 cm every hour; Candle B is 16 cm tall and burns 1 cm every hour. A student claims: *"Candle B will still be burning after Candle A has gone out."*

(a) Write a formula for the height of each candle after t hours.

[1 mark]

(b) Investigate the student's claim. Find when the two candles have the same height, and state what your result means for the claim. Justify fully.

[3 marks]